

Routing Errors Matter: A Lightweight Conditional Hierarchy for Dermoscopic Skin Lesion Classification

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Abstract—Flat multiclass classifiers conceal where errors arise, whereas conditional hierarchies expose screening and subtype decisions but can propagate routing errors. We compare a two-stage conditional hierarchy with a flat four-class classifier using EfficientNet-B0 and the same locked, leakage-aware ISIC 2019 internal-test cohort of 3,668 dermoscopic images. The hierarchy first separates non-malignant from malignant lesions and then conditionally classifies melanoma, basal cell carcinoma, or squamous cell carcinoma. Its accuracy and macro-F1 were 0.740 and 0.605, compared with 0.742 and 0.619 for the flat model. The paired flat-minus-hierarchical macro-F1 difference was 0.014 (95% bootstrap confidence interval: -0.014 to 0.042), and exact McNemar testing did not detect a paired-correctness difference ($p = 0.821$). With oracle routing, hierarchical macro-F1 rose to 0.794, revealing a 0.188 routing-related loss; 20.08% of malignant lesions were blocked at the first stage. A separate five-class melanoma T-category feasibility experiment achieved macro-F1 0.276, with zero recall for T2 and T4. Thus, overall performance was not statistically distinguishable on this split, while routing error was the hierarchy’s main measured limitation and rare T-categories remained inadequately learned.

Index Terms—dermoscopy, hierarchical classification, skin lesions, routing error, class imbalance, EfficientNet

I. INTRODUCTION

Deep neural networks have enabled strong image-based skin-lesion classifiers [?], [?], aided by public dermoscopic collections such as HAM10000 [?] and ISIC challenges [?]. Most systems nevertheless make one flat multiclass decision. A hierarchy can instead represent a clinically meaningful coarse-to-fine decision, but a lesion sent down the wrong branch cannot be recovered by a downstream classifier.

This work studies that failure mode without claiming clinical utility. Its contributions are: (1) a fair same-split comparison of flat and conditional hierarchical EfficientNet-B0 classifiers; (2) paired analysis of their overall performance; (3) oracle-routing decomposition of loss caused by error propagation; and (4) a separately reported melanoma T-category feasibility study under severe label scarcity. We ask whether the hierarchy changes four-class performance and, more importantly, how much of its loss is attributable to routing.

II. RELATED WORK

EfficientNet scales network depth, width, and resolution jointly and offers a compact transfer-learning backbone [?]. Dermoscopic studies have used deep ensembles and metadata to address heterogeneous lesion appearance [?]; controlled

TABLE I
LOCKED FOUR-CLASS EXPERIMENTAL SETUP

Class	Train	Validation	Test
Non-malignant	11,193	2,398	2,398
Melanoma	3,164	678	678
BCC	2,327	498	498
SCC	440	94	94
Total	17,124	3,668	3,668

reader studies also show why evaluation design and population scope matter [?].

Hierarchical prediction incorporates label structure rather than treating all errors identically. Barata *et al.* combined a lesion hierarchy with attention [?], while hierarchical medical-image classifiers have stacked models for coarse and fine decisions [?]. Such factorization creates an error-propagation risk that headline accuracy alone does not reveal. For imbalance, focal loss downweights easy examples [?], and effective-number weighting accounts for diminishing information from repeated samples [?]. We combine these ideas only in the malignant-subtype stage and focus on measured routing effects rather than superiority claims.

III. MATERIALS AND METHODS

A. Dataset and Leakage-Aware Split

We used the public ISIC 2019 training collection, whose sources include HAM10000 and other multi-institutional dermoscopic sets [?], [?]. Diagnoses were mapped to four classes in the fixed order: non-malignant, melanoma, basal cell carcinoma (BCC), and squamous cell carcinoma (SCC). The seed-42 split was frozen before evaluation. Images sharing a lesion identifier or exact file hash were assigned as connected components; records without lesion identifiers used exact-hash grouping when duplicated and otherwise remained singletons. The resulting train, validation, and internal-test sizes were 17,124, 3,668, and 3,668. ISIC 2019 metadata did not provide patient identifiers, so patient-independent splitting could not be guaranteed. Table ?? gives the class distribution.

B. Flat and Conditional Hierarchical Models

Both designs used ImageNet-initialized EfficientNet-B0 [?]. The flat model produced one four-class prediction. In the hierarchy (Fig. ??), Stage 1 classified non-malignant versus

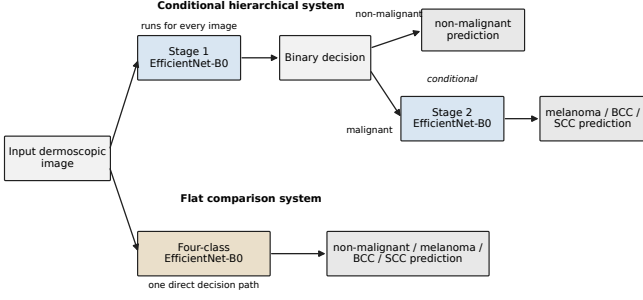


Fig. 1. Conditional hierarchical and flat architectures. Stage 1 runs for every hierarchical input; Stage 2 is conditional on the malignant route.

malignant. An argmax non-malignant decision terminated inference; a malignant decision invoked Stage 2, which classified melanoma, BCC, or SCC. This is a two-stage four-class system. The later T-category study was not integrated into it.

C. Training and Evaluation

Images were ImageNet-normalized; evaluation resized the shorter side to 256 pixels and center-cropped to 224×224 . Moderate training augmentation was fixed across the main comparison. Models were trained for at most 30 epochs with AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), cosine annealing, mixed precision, batch size 64, and early stopping patience 7. Checkpoints were selected only by validation macro-F1.

Stage 1 and the flat model used cross-entropy. Stage 2 used focal loss [?] with $\gamma = 2$ and effective-number class weights [?] computed on its training subset ($\beta = 0.9999$). Each selected checkpoint was evaluated once on the locked test split. We report accuracy, balanced accuracy, macro-F1, and class F1; macro-F1 is primary because SCC is rare.

D. Routing and Statistical Analysis

Predictions were paired by image identifier. A ground-truth-class-stratified paired bootstrap (10,000 replicates, seed 42, percentile 95% intervals) preserved class support and estimated flat-minus-hierarchical differences. An exact two-sided McNemar test used the discordant correctness pairs. An interval including zero is interpreted as failure to establish a statistically distinguishable difference, not equivalence or non-inferiority.

For diagnostic decomposition, oracle gating used the true non-malignant/malignant label while retaining Stage 2 predictions. Its gap from predicted gating quantifies routing-related loss, although the oracle is not a deployable system.

IV. RESULTS

A. Main Four-Class Comparison

Table ?? shows similar point estimates. Flat minus hierarchical macro-F1 was 0.013855 (95% CI: $-0.014255, 0.041963$); accuracy difference was 0.001908 (95% CI: $-0.011996, 0.015812$). The flat model alone was correct on

TABLE II
MAIN COMPARISON ON THE SAME 3,668 IMAGES

Model	Acc.	Bal. acc.	Macro-F1	Weighted-F1
Flat	0.7421	0.6503	0.6192	0.7526
Hierarchy	0.7402	0.6312	0.6054	0.7503
Flat-hier.	0.0019	0.0191	0.0139	0.0022

Macro-F1 difference 95% CI: $[-0.0143, 0.0420]$; exact McNemar $p = 0.8207$.

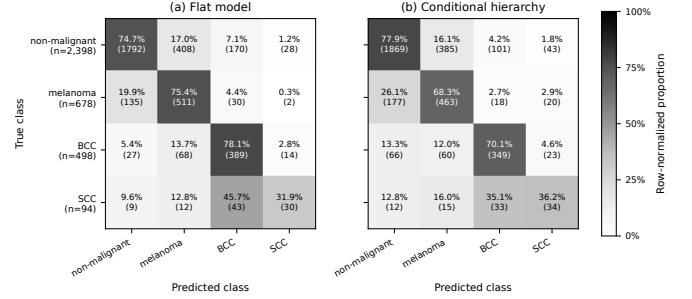


Fig. 2. Row-normalized confusion matrices on the locked test split. Rows are true classes and columns predicted classes; cells also show raw counts.

354 images and the hierarchy alone on 347; exact McNemar $p = 0.820742$. Therefore, neither paired analysis established an overall difference on this split.

B. Routing-Error Decomposition

With oracle routing, four-class macro-F1 was 0.793656 versus 0.605367 with the predicted gate, an absolute loss of 0.188289 (Table ??). Stage 1 blocked 255 of 1,270 malignant images (20.079%) and incorrectly sent 529 of 2,398 non-malignant images to Stage 2 (22.060%). Among 1,015 correctly routed malignant images, 169 (16.650%) still received the wrong subtype. Thus routing was the largest measured source of hierarchical loss, while subtype discrimination remained a secondary limitation.

C. Per-Class Behaviour

Figure ?? displays the error distributions. Flat versus hierarchical F1 was 0.822 versus 0.827 for non-malignant, 0.609 versus 0.578 for melanoma, 0.688 versus 0.699 for BCC, and 0.357 versus 0.318 for SCC. These comparisons are exploratory: no class-wise hypothesis tests or multiplicity adjustment were performed. SCC conclusions are especially unstable because support was only 94.

D. Standalone Stage-3 Feasibility

A separate ISIC-derived melanoma subset contained 848 eligible images labelled from official metadata into broad Tis, T1, T2, T3, and T4 categories. Known patient, lesion, and exact-hash relations were kept within a seed-42 594/127/127 split. Two EfficientNet-B0 candidates differed only in loss and were selected on validation macro-F1 before test access. The preselected inverse-frequency weighted cross-entropy model obtained test macro-F1 0.275611 and balanced accuracy

TABLE III
ROUTING DECOMPOSITION AND STANDALONE FEASIBILITY

Quantity	Value	Denominator
Oracle-gate macro-F1	0.7937	3,668
Predicted-gate macro-F1	0.6054	3,668
Routing-related loss	0.1883	—
Malignant blocked	20.079%	1,270
Non-malignant sent to Stage 2	22.060%	2,398
Stage-3 weighted-CE macro-F1	0.2756	127
Stage-3 balanced accuracy	0.3860	127
Stage-3 T2 / T4 recall	0 / 0	7 / 1

0.386039 (ordinary cross-entropy: 0.162838 and 0.202403). T2 and T4 recall remained zero; their test supports were seven and one. This experiment is feasibility evidence only, not clinical staging or an evaluated third pipeline stage.

V. DISCUSSION AND LIMITATIONS

The hierarchy and flat comparator were not statistically distinguishable in overall performance on this locked split. The oracle analysis adds information that a flat comparison cannot: the hierarchy had a substantially higher diagnostic ceiling, but Stage 1 errors prevented it from approaching that ceiling. Improving the gate could therefore matter more than changing the subtype model, although SCC remained difficult even after correct routing.

Several limitations constrain interpretation. Results come from one seed and one internal ISIC 2019 split; no external evaluation was completed. Known lesion and duplicate relations were leakage-controlled, but absent patient identifiers prevent a guarantee of patient independence. The study did not evaluate calibration, fairness across skin tones, explainability, clinical workflow, latency, energy use, or deployment readiness. Bootstrap inference conditions on the observed cohort, and a confidence interval containing zero does not prove equivalence. Class-wise findings are exploratory, especially for 94 SCC images. Finally, the T-category subset is small, highly imbalanced, metadata-derived, and standalone; its rare-category results are inadequate and it was never evaluated as part of a three-stage pipeline.

VI. CONCLUSION

On the same locked 3,668-image cohort, flat and conditional hierarchical EfficientNet-B0 classifiers had statistically indistinguishable overall performance. Oracle gating exposed a 0.188 macro-F1 routing-related loss and a 20.08% malignant blocking rate, identifying Stage 1 error propagation as the hierarchy’s main measured limitation. A separate T-category study further showed that feasibility does not overcome severe rare-label scarcity. Future work should validate routing improvements and the full system on independent data before any clinical interpretation.